Predicting Credit Card Consumption

A Machine Learning Project for a Retail Banking Use Case

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This document summarizes the problem context, dataset structure, and evaluation approach for an independent project predicting future credit card spend for bank customers using demographic, behavioural, and transactional data.

Background

Banks and card issuers sit on large volumes of behavioural data — transaction history, repayment patterns, account activity, and product holdings. Used well, this data helps an institution understand how a customer is likely to spend in the near future, which in turn supports sharper credit decisions, more relevant marketing, and better-targeted product offers.

This project looks at one specific piece of that puzzle: estimating how much a customer will spend on their credit card over the next few months, based on who they are (demographics) and how they have behaved recently (transactions, balances, loans, and investments).

Objective

Given customer-level demographic and behavioural data, the goal is to build a regression model that predicts average credit card spend for the upcoming three months. A subset of customers in the dataset already have this figure recorded; the model is trained on those records and then used to fill in the missing values for the remaining customers.

Three data sources feed into this problem, joined on a common customer ID:

- Customer demographics — who the customer is
- Customer behaviour — how they transact, save, and borrow
- Credit consumption — the target variable to be predicted

Evaluation Approach

Model quality is assessed using **Root Mean Squared Percentage Error (RMSPE)** between predicted and actual spend, alongside standard regression diagnostics — RMSE and R^2 — to sanity-check overall fit.

Dataset Structure

The three source tables and their fields are summarized below. All tables share a common **ID** column used to join them into a single customer-level view.

1. Customer Demographics

Field	Description
ID	Unique identifier for each customer
Account_type	Type of bank account held (current or savings)
Gender	Customer's gender (M or F)
Age	Customer's age
Income	Income bracket (High / Medium / Low)
Emp_Tenure_Years	Years of employment experience

Field	Description
Tenure_with_Bank	Number of years the customer has banked with the institution
Region_code	Ordered code representing the customer's region of residence
NetBanking_Flag	Whether the customer actively uses net banking
Avg_days_between_transaction	Average gap, in days, between consecutive transactions

2. Customer Behaviour

Field	Description
ID	Unique identifier for each customer
CC_cons_apr / may / jun	Credit card spend for April, May, and June respectively
DC_cons_apr / may / jun	Debit card spend for April, May, and June respectively
CC_count_apr / may / jun	Number of credit card transactions per month
DC_count_apr / may / jun	Number of debit card transactions per month
Card_lim	Maximum credit card limit assigned to the customer
Personal_loan_active	Whether an active personal loan exists with another bank
Vehicle_loan_active	Whether an active vehicle loan exists with another bank
Personal_loan_closed	Whether a personal loan was closed in the last 12 months
Vehicle_loan_closed	Whether a vehicle loan was closed in the last 12 months
Investment_1	DEMAT account investment value in June
Investment_2	Fixed deposit investment value in June
Investment_3	Life insurance investment value in June
Investment_4	General insurance investment value in June
Debit_amount_apr / may / jun	Total amount debited from the account each month
Credit_amount_apr / may / jun	Total amount credited to the account each month
Debit_count_apr / may / jun	Number of debit transactions on the account each month
Credit_count_apr / may / jun	Number of credit transactions on the account each month
Max_credit_amount_apr / may / jun	Largest single credit amount received each month

Field	Description
Loan_enq	Whether the customer made a loan enquiry in the last 3 months (Y/N)
Emi_active	Monthly EMI amount paid toward active loans with other lenders

3. Credit Consumption (Target)

Field	Description
ID	Unique identifier for each customer
cc_cons	Target variable: average credit card spend over the next three months. Missing for the customers the model is ultimately used to predict for.

Note: A portion of customers have no recorded value for cc_cons. The model is trained on customers with a known value and then applied to estimate spend for the remaining customers.

Deliverables

The completed project covers the following pieces:

- End-to-end, commented code covering data preparation through prediction
- Exploratory data analysis of the merged customer dataset
- Model training, validation, and comparison across algorithms
- Documentation of modelling choices and evaluation results
- Final predicted spend values for customers with missing targets